DOOB'S PROOF OF MLE CONSISTENCY

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Abstract. We present a concise exposition of Doob's proof of MLE consistency using surprisal and relative information. We clarify historical entropy terminology and explicitly verify all assumptions for the multivariate normal distribution using moment-generating functions and elementary matrix algebra.

1. Introduction

The consistency of Fisher's maximum likelihood estimator (MLE) [10] is a basic result discovered more than a century ago. This method is typically treated as a special case of M-estimators, and derived via standard techniques dating back to Cramér [6] and Wald [26]. A modern treatment is van der Vaart [25]. Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found Doob's consistency proof [9] of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter spaces and general state spaces with no extra effort only adds to its appeal.

The contribution of this note is expository: it gives a formulation of Doob's consistency argument in surprisal/information terminology, emphasizing the role of properness and separability, and works out the multivariate normal example in detail. The proofs are designed to be self-contained throughout: in particular, the consistency argument and the matrix lemmas are developed directly, without relying on determinant multiplicativity, eigenvalue continuity, or other standard results that can obscure the elementary structure of the argument.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2. Surprisal and Information

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X. We assume the density $p(X, \theta^*)$ of X is one of a parametric family $p(X, \theta)$ of positive densities, and we wish to recover θ^* from the samples $X_1, X_2, \dots$.

Stated in this manner, one may interpret the family $p(X, \theta)$ as a probe of the underlying expectation E_*. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over E_*. Given this, how do we use the samples $X_1, X_2, \dots$ to recover θ^*?

Define the *surprisal* [19, 23]

$$S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to call a minimizer θ^* of the *expected surprisal* $E_*(S(X, \theta))$ a *true parameter*.

Equivalently, given θ^*, the *relative information* is

$$I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)). \tag{1}$$

The relative information is the excess expected surprisal of θ over θ^*. Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ. The relative information I appears in many fields, in different forms. Because of widely differing terminology, below we give an account of its history.

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While $p(X,\theta)$ have so far been arbitrary positive functions, we now specialize to the case where $p(X,\theta)$ are densities in the usual sense. We say θ^* is a *reference parameter* if there exists a background expectation E such that

$$E_*(f(X)) = E(p(X,\theta^*)f(X)) \qquad \text{and} \qquad E_\theta(f(X)) = E(p(X,\theta)f(X)) \tag{2}$$

hold, with $E_\theta(1) = 1$ for all θ. Equivalently, θ^* is a reference parameter if

$$E_*\left(\frac{p(X,\theta)}{p(X,\theta^*)}\right) = 1 \qquad \text{for all } \theta. \tag{3}$$

When θ^* is a reference parameter, the relative information takes the familiar form [7, 8] (see §4 for terminology)

$$I(\theta^*,\theta) = E\left(p(X,\theta^*)\log\left(\frac{p(X,\theta^*)}{p(X,\theta)}\right)\right). \tag{4}$$

Since $j(p) = \log(1/p)$ is convex, by Jensen's inequality,

$$I(\theta^*,\theta) = E_*\left(j\left(\frac{p(X,\theta)}{p(X,\theta^*)}\right)\right) \geq j\left(E_*\left(\frac{p(X,\theta)}{p(X,\theta^*)}\right)\right) = j(1) = 0, \qquad \text{for all } \theta.$$

It follows that a reference parameter is a true parameter. If θ is any other true parameter, then $I(\theta^*,\theta) = 0$, hence, by strict convexity of $j(p)$, there is a scalar t such that $p(X,\theta) = t \cdot p(X,\theta^*)$. Since θ^* is a reference parameter, $t = 1$, hence $p(X,\theta^*) = p(X,\theta)$.

We say parameters θ, θ' *share a density* if $p(X,\theta) = p(X,\theta')$. It follows that reference parameters share a density, and true parameters share a density when there is a reference parameter. From this, there is at most one reference parameter when parameters do not share densities; when there is a reference parameter, the true parameter is the reference parameter, when parameters do not share densities.

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of X is $p(X,\theta^*) = \theta_X^*$. If θ is any other probability vector, the surprisal is $S(X,\theta) = \log(1/\theta_X)$, and

$$I(\theta^*,\theta) = \sum_{i=1}^{d} \theta_i^* \log(\theta_i^*/\theta_i). \tag{5}$$

Turning back to the general setting, the *likelihood* and (empirical) *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^{n} p(X_k,\theta), \qquad \bar{S}_n(\theta) = \frac{1}{n}\log(1/L_n(\theta)) = \frac{1}{n}\sum_{k=1}^{n} S(X_k,\theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes $E_*(S(X,\theta))$. By the (strong) law of large numbers (LLN), $\bar{S}_n(\theta)$ converges to $E_*(S(X,\theta))$ as $n \to \infty$, with probability one, for each θ. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ. In what follows, only E_*, $p(X,\theta)$, and true parameters enter; E, E_θ, and reference parameters play no role.

3. Consistency

Our first assumption is

A. *(Identifiability):* For every θ, $S(X,\theta)$ is integrable and there is at most one true parameter.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c. Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the second assumption

B. *(Properness):* There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,

$$F(\theta) \leq \bar{S}_n(\theta), \qquad \text{for all } \theta \text{ and } n \geq N. \tag{6}$$

When the parameter space is compact, every continuous function $F(\theta)$ is proper, and the proof below shows assumption **B** is superfluous. We will need the LLN convergence $\bar{S}_n(\theta) \to E_*(S(X,\theta))$ across θ, in the sense

$$\bar{S}_n(\theta) \to E_*(S(X,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta, \tag{7}$$

with probability one. For this we need the third assumption

C. *(Continuity):* The parameter space is separable and there is a continuous $\delta(\theta,\theta')$ and an integrable nonnegative $g(X)$ satisfying

$$|S(X,\theta) - S(X,\theta')| \le \delta(\theta,\theta')g(X) \qquad \text{and} \qquad \delta(\theta,\theta) = 0. \tag{8}$$

Under **C**, a standard separability argument, given in the appendix, establishes (7) with probability one. Under **A** and **C**, the expected surprisal $E_*(S(X,\theta))$ is continuous in θ. If moreover **B** holds, then by sending $n \to \infty$ in (6), $F(\theta) \le E_*(S(X,\theta))$ for all θ, hence the expected surprisal is proper. Since a proper continuous function has a minimizer, under **A**, **B**, and **C**, there is a unique true parameter. Similarly, under **B** and **C**, the mean surprisal is continuous in θ and proper, hence an MLE $\hat{\theta}_n$ exists for $n \ge N$, with probability one.

These assumptions are particularly tractable for *exponential families* [5, 16, 20, 25], the densities $p(X,\theta)$ where $S(X,\theta)$ is a sum of products of the form $f(\theta)g(X)$.

In (7), there is no presumption of uniform convergence. This is a key feature of Doob's proof: no uniform convergence of the mean surprisal is required. The argument instead combines properness, the LLN, and the continuity estimate in (8) to control limit points of the MLE sequence.

Theorem (Consistency)**.** *Let $X_1, X_2, \dots$ be independent and identically distributed, and let $p(X,\theta)$ be a parametric family of positive functions. Under the above assumptions,*

(1) the expected surprisal is proper and there is a unique true parameter θ^*,
(2) with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n,
(3) with probability one, any MLE sequence converges to the true parameter, $\hat{\theta}_n \to \theta^*$.

Proof. The first two items were derived above. For the third item, by the LLN,

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to E_*(g(X)), \qquad \text{as } n \to \infty, \tag{9}$$

with probability one. Fix samples for which (6), (7) and (9) hold. We show $\hat{\theta}_n \to \theta^*$ for these samples. By (6),

$$F(\hat{\theta}_n) \le \bar{S}_n(\hat{\theta}_n) \le \bar{S}_n(\theta^*), \qquad \text{for } n \ge N.$$

Since $\bar{S}_n(\theta^*) \to E_*(S(X,\theta^*))$, the right side is bounded. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space, and thus has limit points.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n \to \theta$ along a subsequence. By (8),

$$\bar{S}_n(\theta) - \bar{S}_n(\theta^*) \le \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \le \delta(\theta,\hat{\theta}_n) \cdot \frac{1}{n}\sum_{k=1}^{n} g(X_k). \tag{10}$$

Since the left side of (9) is bounded as $n \to \infty$, and $\delta(\theta,\hat{\theta}_n) \to 0$ along a subsequence, the right side of (10) goes to zero along a subsequence. By (1) and (7), the left side of (10) converges to $I(\theta^*,\theta)$, hence $I(\theta^*,\theta) \le 0$. But $I(\theta^*,\theta) \ge 0$, hence $I(\theta^*,\theta) = 0$, showing $\theta = \theta^*$. Thus $\hat{\theta}_n \to \theta^*$ as $n \to \infty$. □

4. Terminology History

The relative information $I(\theta^*,\theta)$ is often called "relative entropy" [5, 16]. The term "entropy," originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann's 1877 combinatorial interpretation of entropy [3] and Planck's 1901 paper [18], reversed Boltzmann's sign convention, writing entropy with the sign convention used today.

Shannon, in his 1948 paper [21], independently defined

$$H(p) = -\sum_{i=1}^{d} p_i \log p_i.$$

Von Neumann had earlier defined quantum entropy in his 1932 book [17]. According to a widely repeated account attributed to Shannon and reported in [24], von Neumann suggested the name "entropy" when Shannon was casting about for terminology after developing his communication theory in the mid-1940s; later [22] Shannon expressed doubt that the conversation took place. Whatever the origin of the name, in his 1948 paper, Shannon explicitly links his H to Boltzmann's: "H is then, for example, the H in Boltzmann's H theorem," connecting his terminology to the earlier usage despite the difference in sign conventions.

The relative information $I(\theta^*, \theta)$ (4) is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, as is apparent in the special case (5), whereas the entropy $H(p)$ above is strictly concave in p. From this perspective, the terminology "relative entropy" can seem counterintuitive. This terminology extends into the Python `scipy.stats` package: `entropy(p)` returns the concave H, whereas `entropy(p,q)` returns the convex I. The issue is the conceptual opposition between entropy and information: entropy is concave, whereas information is convex.

Indeed, based on curvature, one might argue that the *negative* of (5) is the more natural candidate for "relative entropy." Similar remarks apply to the "cross-entropy" loss J of machine learning. Since J differs from I by a constant, one might analogously call J the *cross-information*, and its negative "cross-entropy."

While many authors use terminology different from ours, some align partially with our usage. Schervish, in his 1995 book [20], uses "Kullback-Leibler information" for I, which is more descriptive than the widely used "Kullback-Leibler divergence" [1, 15]. In particular, Kullback himself, in his 1959 monograph [14], refers to I as an "information function." Donsker-Varadhan, in their 1975 large deviations paper [8], denote their rate function by I rather than H; the relative information is the i.i.d. special case of their I.

Doob's 1934 proof appeared at the dawn of modern probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of the Kolmogorov extension theorem [13].

5. Multivariate Normal

Let μ be in $\mathbf{R}^d$, Q a positive symmetric $d \times d$ matrix, and $\theta = (\mu, Q)$. Let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(X, \theta) = e^{-S(X,\theta)}$ where

$$S(X, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2}(X - \mu) \cdot Q^{-1}(X - \mu). \tag{11}$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter $\theta^* = (\mu^*, Q^*)$* if its moment-generating function is

$$E_* \left(e^{v \cdot X} \right) = \exp \left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v \right). \tag{12}$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (12), and let X_1, X_2, $\ldots$ be independent and identically distributed copies[1] of X. We show θ^* is the true parameter, and we verify **A**, **B**, and **C**.

Assumption **C** with $g(X) = 1 + |X|^2$ follows from (11). To establish **A** and **B**, we review basic matrix facts [12], stated as lemmas and derived in the appendix. The proofs deliberately avoid invoking determinant multiplicativity or continuity properties of eigenvalues: the point is not merely to verify the required facts, but to expose the elementary mechanism behind them.

We say a symmetric matrix Q is *positive*, and we write $Q > 0$, if $v \cdot Qv > 0$ for $v \neq 0$. We say a symmetric matrix Q is *nonnegative*, and we write $Q \geq 0$, if $v \cdot Qv \geq 0$ for all v. Given symmetric matrices Q, R we write $Q \geq R$ if $Q - R \geq 0$. If $Q > 0$, then $Q \geq \delta I$ for some $\delta > 0$. If $Q \geq 0$ and $R \geq S$, then $\operatorname{trace}(QR) \geq \operatorname{trace}(QS)$. Moreover $\operatorname{trace}(QR) = \operatorname{trace}(RQ)$ and we have the Cauchy-Schwarz inequality

$$\operatorname{trace}(QR) \leq \sqrt{\operatorname{trace}(Q^2)} \sqrt{\operatorname{trace}(R^2)}, \qquad Q, R \text{ symmetric}.$$

[1] This setup is a special case of the Kolmogorov extension theorem.

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v \otimes w$ is the matrix whose ij-th entry is $v_i w_j$. Then $v \otimes v$ is nonnegative and

$$v \cdot Qv = \mathrm{trace}(Q(v \otimes v)).$$

Let $R = v \otimes v$. Since $R^2 = |v|^2 R$ and $\mathrm{trace}(R) = |v|^2$, $\mathrm{trace}(R^2) = |v|^4$. Using this and Cauchy-Schwarz,

$$\delta^2 = \mathrm{trace}(Q^2) \qquad \Longrightarrow \qquad Q \le \delta I. \tag{13}$$

If $u = \sqrt{\epsilon} v - w/\sqrt{\epsilon}$, then $u \otimes u \ge 0$, hence

$$v \otimes w + w \otimes v \le \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w. \tag{14}$$

If Q_n is a sequence of symmetric matrices with $Q_n \to 0$, then $\mathrm{trace}(Q_n^2) \to 0$, hence by (13), given $Q > 0$, there is $N \ge 1$ such that $Q_n \le Q$ for $n \ge N$.

By the eigenvalue decomposition, every nonnegative matrix is a sum of matrices of the form $v \otimes v$, and every positive matrix is a sum of δI and matrices of the form $v \otimes v$.

The determinant $\det(Q)$ is the product of its eigenvalues.

Lemma 1. *With Q invertible symmetric and t a scalar,* $\det(Q + tv \otimes v) = \det(Q)\left(1 + tv \cdot Q^{-1} v\right)$.

Lemma 2. *If $S > 0$ and $Z > 0$, then* $\det(SZS) = \det(Z)\det(S)^2$.

Lemma 3. *For c scalar, let $\epsilon = e^{-c}$ and $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$, and $h(\lambda) \le c$ implies $\epsilon \le \lambda \le 1/\epsilon$.*

Lemma 4. *For $Q^* > 0$ and c scalar, let $\epsilon = e^{-2c-1}\det(2\pi e Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2}\log\det(2\pi Q) + \frac{1}{2}\mathrm{trace}(Q^{-1}Q^*), \qquad Q > 0. \tag{15}$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \le c$ implies $\epsilon Q^* \le Q \le Q^*/\epsilon$.

Let $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$. Given $\theta = (\mu, Q)$, let

$$F(\mu, Q) = F_*(Q, f(\mu)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

By Lemma 4, (μ^*, Q^*) is the unique minimizer of $F(\mu, Q)$.

In (12), multiply both sides by $e^{-v\cdot\mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$E_*(X) = \mu^*, \qquad E_*\left((X - \mu) \otimes (X - \mu)\right) = f(\mu).$$

It follows $E_*(S(X, \theta)) = F(\mu, Q)$. By Lemma 4, we have **A** and $\theta^* = (\mu^*, Q^*)$ is the true parameter.

The second consequence of Lemma 4 is properness of $F(\mu, Q)$. If $F(\mu, Q) \le c$, then $F_*(Q, Q^*) \le c$, hence $\epsilon Q^* \le Q \le Q^*/\epsilon$, confining Q to a compact subset of the space of positive definite matrices. From this,

$$\begin{aligned} c &\ge F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu) \\ &\ge F_*(Q^*, Q^*) + \frac{\epsilon}{2}(\mu^* - \mu) \cdot (Q^*)^{-1}(\mu^* - \mu), \end{aligned}$$

confining μ to a compact subset of $\mathbf{R}^d$. Thus $F(\mu, Q)$ is proper.

Let

$$f_\epsilon(\mu) = (1 - \epsilon)f(\mu) - \epsilon Q^*, \qquad F_\epsilon(\theta) = F_\epsilon(\mu, Q) = F_*(Q, f_\epsilon(\mu)).$$

Since $f_\epsilon(\mu) \ge (1 - 2\epsilon)f(\mu)$,

$$F_\epsilon(\mu, (1 - 2\epsilon)Q) \ge \frac{1}{2}\log\det((1 - 2\epsilon)I) + F(\mu, Q),$$

hence $F_\epsilon(\theta)$ is proper.

With

$$Q_n(\mu) = \frac{1}{n}\sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu), \qquad \hat{\mu}_n = \frac{1}{n}\sum_{k=1}^n X_k, \qquad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$, and let $0 < \epsilon < 1/2$.

By the LLN, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \to 0, \qquad Q^* - Q_n(\mu^*) \to 0, \qquad \text{as } n \to \infty.$$

Thus there is $N \geq 1$, depending on the samples, such that, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon^2 Q^*, \qquad Q_n(\mu^*) \geq (1-\epsilon)Q^*, \qquad \text{for } n \geq N. \tag{16}$$

Fix samples for which (16) holds. We establish (6) for these samples and $F(\theta) = F_\epsilon(\theta)$.

With $w = \hat{\mu}_n - \mu^*$ and $v = \mu - \mu^*$, using (14),

$$v \otimes w + w \otimes v \leq \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon f(\mu)$$

for all μ and $n \geq N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$Q_n(\mu) = Q_n(\mu^*) - (v \otimes w + w \otimes v) + v \otimes v.$$

Combined with the above inequalities, this leads to

$$Q_n(\mu) \geq f_\epsilon(\mu), \qquad \text{for all } \mu \text{ and } n \geq N.$$

In particular, since $f_\epsilon(\mu) \geq (1-2\epsilon)Q^*$, we have shown $\hat{Q}_n > 0$ for $n \geq N$.

Since $\bar{S}_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \geq \hat{Q}_n$ by least squares, by Lemma 4,

$$\bar{S}_n(\theta) = F_*(Q, Q_n(\mu)) \geq F_*(Q, \hat{Q}_n) \geq F_*(\hat{Q}_n, \hat{Q}_n) = \bar{S}_n(\hat{\theta}_n),$$

for all θ and $n \geq N$. Thus $\hat{\theta}_n$ is the MLE for $n \geq N$, and

$$\bar{S}_n(\theta) = F_*(Q, Q_n(\mu)) \geq F_*(Q, f_\epsilon(\mu)) = F_\epsilon(\theta), \qquad \text{for all } \theta \text{ and } n \geq N.$$

This establishes (6) and completes the derivation of **B**.

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Appendix A. Auxiliary Proofs

Proof of (7) with probability one. By separability of the parameter space, there is a countable dense set D of parameters. Since D is countable, with probability one,

$$\bar{S}_n(\theta') \to E_*(S(X, \theta')), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta' \text{ in } D. \tag{17}$$

Fix samples for which (9) and (17) hold. We establish (7) for these samples. If θ, θ' are any parameters, by (8) and the triangle inequality,

$$\begin{aligned} |\bar{S}_n(\theta) - E_*(S(X,\theta))| \leq & |\bar{S}_n(\theta') - E_*(S(X,\theta'))| \\ & + \delta(\theta, \theta') \left(E_*(g(X)) + \frac{1}{n}\sum_{k=1}^n g(X_k) \right). \end{aligned} \tag{18}$$

Now take the $\limsup_{n\to\infty}$ in (18). By (9) and (17),

$$\limsup_{n\to\infty} |\bar{S}_n(\theta) - E_*(S(X,\theta))| \leq 2\delta(\theta,\theta')E_*(g(X)), \quad \text{for all } \theta' \text{ in } D \text{ and all } \theta.$$

Since D is dense, $\delta(\theta, \theta')$ is continuous, and $\delta(\theta,\theta) = 0$, for fixed θ and varying θ', the right side is arbitrarily small, while the left side does not depend on θ'. This establishes (7). □

Proof of Lemma 1. Assume $t \neq 0$. If v is orthogonal to an eigenvector x of Q, then we reduce to a lower-dimensional case by restricting to $x^\perp$. Hence we may assume $v \cdot x \neq 0$ for all eigenvectors x of Q. If v is orthogonal to an eigenvector x' of $Q' = Q + tv \otimes v$, then x' is an eigenvector of Q, contradicting our assumption. Hence we may assume $v \cdot x' \neq 0$ for all eigenvectors x' of Q'. If x'_1 and x'_2 are linearly independent eigenvectors of Q' corresponding to the same eigenvalue, then, for suitable r_1, r_2, $x' = r_1 x'_1 + r_2 x'_2$ is an eigenvector of Q' satisfying $v \cdot x' = 0$, contradicting our assumptions. Hence we may assume the eigenvalues of Q' are simple. If x and x' are eigenvectors for Q and Q' corresponding to the same eigenvalue, then

$$t(v \cdot x')(v \cdot x) = x' \cdot (tv \otimes v)x = x' \cdot Q'x - x' \cdot Qx = 0,$$

contradicting our assumptions. Hence we may assume the eigenvalues of Q and the eigenvalues of Q' do not overlap. Let λ_1, λ_2, ..., λ_d be the eigenvalues of Q and λ'_1, λ'_2, ..., λ'_d the eigenvalues of Q'. Let v_1, v_2, ..., v_d be the corresponding orthonormal basis of eigenvectors of Q. Then the *secular equation* is

$$\sigma(z) = 1 + tv \cdot (Q - zI)^{-1} v = 1 + t \sum_{i=1}^{d} \frac{(v \cdot v_i)^2}{\lambda_i - z} = 0.$$

If λ' is an eigenvalue of Q' with eigenvector x', then $(Q' - \lambda' I)x' = 0$, hence $(Q - \lambda' I)x' + t(v \cdot x')v = 0$. Taking the dot product with $(Q - \lambda' I)^{-1} v$ yields $v \cdot x' + t(v \cdot x')(v \cdot (Q - \lambda' I)^{-1} v)$, hence $\sigma(\lambda') = 0$. If we let

$$\det(Q - zI) = \prod_{i=1}^{d} (\lambda_i - z), \qquad p(z) = \det(Q - zI)\sigma(z),$$

then $p(z)$ is a polynomial and $p(\lambda') = 0$. Since λ'_1, λ'_2, ..., λ'_d are simple, $p(z) = \prod_{i=1}^{d} (\lambda'_i - z)$. Since the product of the roots is $p(0)$, the result follows. □

Proof of Lemma 2. Assume the result holds for Z. By Lemma 1,

$$\begin{aligned}\det(S(Z + v \otimes v)S) &= \det(SZS + Sv \otimes Sv) \\ &= \det(SZS)(1 + Sv \cdot (SZS)^{-1} Sv) \\ &= \det(Z)\det(S)^2(1 + v \cdot Z^{-1} v) = \det(Z + v \otimes v)\det(S)^2,\end{aligned}$$

hence the result holds for $Z + v \otimes v$. If $Z = \delta I$, the result holds. Since every $Z > 0$ is a sum of δI and a sum of matrices of the form $v \otimes v$, the result follows by induction over the number of terms in the sum. □

Proof of Lemma 3. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \le c$ implies $\lambda \le 1/\epsilon$. Since $\lambda \log \lambda \ge -1/e$ and $(e-1)e^c \ge ec$ for $c \ge 1$, $h(\lambda) \le c$ and $c \ge 1$ implies $\lambda \ge (1 - 1/e)/c \ge \epsilon$. □

Proof of Lemma 4. Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1} Q S^{-1}$ and let λ_1, λ_2, ..., λ_d be the eigenvalues of Z. Then $\mathrm{trace}(Q^{-1} Q^*) = \mathrm{trace}(S Q^{-1} S) = \mathrm{trace}(Z^{-1})$. By Lemmas 2 and 3,

$$\begin{aligned}2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) &= \log\det Q - \log\det Q^* + \mathrm{trace}(Q^{-1} Q^* - I) \\ &= \log\det(SZS) - \log\det(S)^2 + \mathrm{trace}(Z^{-1} - I) \\ &= \log\det Z + \mathrm{trace}(Z^{-1} - I) \\ &= \sum_{i=1}^{d} (h(\lambda_i) - 1) \ge 0, \qquad (19)\end{aligned}$$

establishing $F_*(Q, Q^*) \ge F_*(Q^*, Q^*)$. If (19) is an equality, $Z = I$ hence $Q = Q^*$. Now suppose $F_*(Q, Q^*) \le c$. Then

$$2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) \le 2c - \log\det(2\pi Q^*) - \mathrm{trace}(I) = \log(1/\epsilon) - 1.$$

Since the terms in the sum in (19) are nonnegative, $h(\lambda_i) - 1 \le \log(1/\epsilon) - 1$ for $i = 1, 2, \ldots, d$. By Lemma 3, $\epsilon \le \lambda_i \le 1/\epsilon$ for $i = 1, 2, \ldots, d$. This implies $\epsilon I \le Z \le I/\epsilon$, hence $\epsilon Q^* \le Q \le Q^*/\epsilon$. □

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